

Memory models for microprocessor 8086

- model small (most common)

code size - 64K

Data size - 64K

- model medium

code size - 64K +

Data size - 64K

- model compact

code - 64K

Data - 64K +

Basic Structure of Assembly language program

- MODEL SMALL

- STACK 100H stack size, 00100h bytes
if not mentioned, 1KB

- DATA

{ data segment }

- CODE

{ code segment }

MAIN ENDP

END MAIN

RET

Data Segment Definition

db → byte

dw → word (2 bytes)

equ → numeric const to name.

var

data_byte DB 255 (1 byte)

data_word DW 512 (2 byte)

data_string DB "Hello"

max_int equ 65535

defined

Addressing modes

- Addressing Data

- Program Codes in memory — used with JMP and CALL

- Stack in memory — push pop

(slide at a glance)

- I/O — IN AL, 05h OUT DL, AL } I/O port num & address

- Implied addressing

only DL register for I/O port

no address given

CLC, HLT, RET

[Write short notes — past yr question]

Addressing Data

Addressing Mode	Description	Example Instruction	Effective Address / Operation
Immediate Addressing	Data is part of the instruction itself	MOV BL, 44 MOV BX, 256	Operand value is used directly
Direct Addressing	Memory address is directly specified in the instruction	MOV AX, [1234H] MOV AX, DATA_VAR	$AX \leftarrow (DS \times 10H + 1234H)$
Register (Direct) Addressing	Operand is located in a register	MOV AX, BX MOV AL, BL	$AX \leftarrow BX$
Register Indirect Addressing	Register holds the memory address of the operand	MOV AX, [BX]	$AX \leftarrow (DS \times 10H + BX)$
Base-Plus-Index Addressing	Base register (BX/BP) + Index register (SI/DI)	MOV DX, [BX+DI]	$DX \leftarrow (DS \times 10H + BX + DI)$
Register Relative Addressing	Base or index register + displacement	MOV DX, [BX+2]	$DX \leftarrow (DS \times 10H + BX + 2)$
Base-Relative-Plus-Index Addressing	Base + Index + displacement	MOV DX, [BX+DI+10]	$DX \leftarrow (DS \times 10H + BX + DI + 10)$

Write an assembly language program structure to allocate exactly 64 Kbytes of memory for code segment, 512 bytes for stack segment and also consider that the size for data segment may exceed 64 Kbytes.

```
.MODEL COMPACT      ; Code can be 64 KB, Data may exceed 64 KB
```

```
.STACK 200H          ; 512 bytes stack (200H = 512 decimal)
```

```
.DATA                ; Data segment (may exceed 64 KB)  
    ; Data declarations  
    VAR1 DW 1234H
```

```
.CODE                ; Code segment (64 KB)
```

```
MAIN PROC
```

```
    MOV AX, @DATA    ; Initialize data segment  
    MOV DS, AX
```

```
    ; Program instructions
```

```
    ; Program termination  
    MOV AH, 4CH  
    INT 21H
```

```
MAIN ENDP
```

```
END MAIN
```

```
RET
```